

Project 2

Image Editor

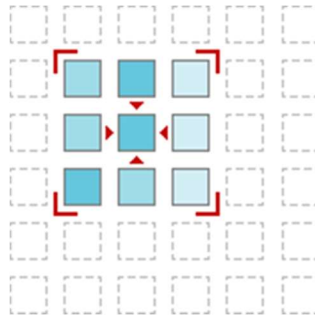
Function: void blur();

Purpose:

In this function you have to blur the input image.

Method:

For this we use average method. Now the question is what is average method? For each index of a 2-D Matrix you have to calculate the average of its neighbor indexes. Each index have 8 neighbor in case of 3x3 average (**not boundary cases**). Your task is to calculate the average value of those 9 index (**8 neighbors and 1 itself**) and place the average value on output matrix.



How to handle boundary cases:

As in boundary cases we should extend our input image as well. To do that we are to add extra rows at the top and at the bottom of the image and add extra columns to the left and to the right. Which contain **value 0**. Number of extra row and column depends on the size of average scale.

Example 3x3:

We add 1 extra row at the top and at the bottom of the image and 1 extra columns to the left and to the right.

Example 5x5:

We add 2 extra row at the top and at the bottom of the image and 2 extra columns to the left and to the right.

Function: void crop();

Purpose:

In this function you have to crop the image.

Method:

Get input from user from where to where user want to crop the image. Now verify the input if input is in given range (**between row and column size of image**) than crop the image according to input. The output image will be smaller in size (**row and column**).

Function: void addborder();

Purpose:

In this function you have to add border to the image.

Method:

We should extend our output image. To do that we are to add extra rows at the top and at the bottom of the image and add extra columns to the left and to the right. Which contain **value 0**. The **more** number of extra rows and columns creates the **thicker** border.

Function: void zoom();

Purpose:

In this function you have to zoom in or zoom out the image.

Method:

For this we use **nearest neighbor** method. For zoom in we **replicate** the image index. And for zoom out we pick **1 value** between **4 values**.

Example zoom in:

		1	1	2	2
		1	1	2	2
		3	3	4	4
		3	3	4	4

1	2
3	4

Example zoom out:

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

1	3
9	11

Function: void removeColor();

Purpose:

In this function we convert color image into gray scale image.

Method:

For this we use **average** method. Average method is the simplest one. You just have to take the average of three colors. Since it's an RGB image, so it means that you have add the values of **r** with **g** with **b** and then divide it by **3** to get your desired grayscale image value.